Fiscal Year:	FY 2015	**Task Last Updated:**	FY 07/01/2015
PI Name:	Derby, Jeffrey J. Ph.D.		
Project Title:	Modeling of Particle Transport in the Melt and Its Interaction with the Solid-Liquid Interface		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	55455-0132	**Congressional District:**	5
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2009 ESA AO Biology on Sounding Rockets and Physical Sciences on All Platforms
Start Date:	09/01/2010	**End Date:**	11/30/2015
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA MSFC
Contact Monitor:	Downey, James1 Patton	**Contact Phone:**	
Contact Email:	patton.downey@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	NOTE: End date is 11/30/2015, per NSSC information (Ed., 6/3/15) NOTE: End date is 8/31/2015, per NSSC information (Ed., 6/3/14) NOTE: End date is 8/31/2014, per NSSC information (Ed., 5/30/2013)		
Key Personnel Changes/Previous PI:	Jeffrey Derby is the PI of this NASA project. Collaborators funded by the DLR ParSiWal project are Jochen Friedrich (Fraunhofer IISB), Arne Croell (University of Freiburg), and Heike Emmerich (University of Bayreuth).		
COI Name (Institution):			
Grant/Contract No.:	NNX10AR70G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The primary goal of this research project is to understand the mechanisms that determine the dynamics of particle transport in silicon melts and the pushing or engulfment of these particles at a moving solid-liquid interface, specifically small silicon carbide and silicon nitride particles during the solidification of silicon. This understanding is vital to improving the quality of solar-grade, multi-crystalline silicon grown by directional solidification processes. This work will coordinate with the ParSiWal (Partikeleinfang bei der Siliziumkristallisation im Weltall) project funded by the German DLR (Deutsches Zentrum für Luft- und Raumfahrt) and involving Jochen Friedrich (Fraunhofer IISB), Arne Croell (University of Freiburg), and Heike Emmerich (University of Bayreuth). Specifically, finite-element, moving-boundary analyses are being developed and applied to assess the dynamics of particle pushing and engulfment. Of particular interest is to ascertain critical velocities for engulfment for a variety of different particle sizes and thermal gradients. We will also study the effects of shear and solidification flows, in the melt near the interface, on engulfment mechanisms and the effect of a grain boundary at the interface that locally modifies its shape and energy. Planned experiments by ParSiWal in the laboratory as well as during the TEXUS mission are aimed to probe the effects of particle engulfment during silicon solidification and will be used to assess theoretical predictions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Understanding the dynamics of particle pushing and engulfment will enable process development to minimize their effects on grown material. This will promote lower cost and higher quality of solar-grade, multi-crystalline silicon and will advance progress in commercial solar cell production.
Task Progress:	Model development is now basically complete, and we are pursuing several sets of interesting computations. We have found that the model predicts significant departures from prior, idealized analytical analyses under certain conditions. A paper is being prepared to report our progress in a suitable, peer-reviewed journal. The submission of this paper is imminent, within the next few weeks. We are also preparing a set of computations that will be specifically focused on silicon carbide particles in solidifying silicon. We have found that, consistent with the statements above, our model is providing a much deeper understanding of engulfment phenomena and is much more accurately representing initial engulfment data taken in ground experiments conducted at the University of Freiburg. A second paper, based on these findings, is being prepared. We have found that our work is being well received by the crystal growth community, and we have made numerous presentations at several important meetings during this past period (see below). We have also submitted abstracts for presentations at future meetings: the 31st Annual Meeting of the American Society for Gravitational and Space Research, Alexandria, VA, November 11–14, 2015, and the TMS Annual Meeting & Exhibition, Nashville, TN, February 14–18, 2016. Supported graduate student Yutao Tao gave two oral presentations: at the 30th Annual Meeting of the American Society for Gravitational and Space Research, Pasadena, CA, October 25, 2014 and at the 20th American Conference on Crystal Growth and Epitaxy (ACCGE-20) and 17th U.S. Biennial Workshop on Organometallic Vapor Phase Epitaxy (OMVPE-17), Big Sky, MT, August 2–7, 2015. PI Derby gave an invited lecture on the topic of this project: “Particle Engulfment Kinetics during the Growth of Crystalline Silicon,” Invited lecture, Kinetic Seminar, Materials and Process Simulation, University of Bayreuth, Bayreuth, Germany, May 4–5, 2015. Derby also gave a contributed presentation on this project at CSSC-8, 8th International Workshop on Crystalline Silicon for Solar Cells, Bamberg, Germany, May 5–8, 2015. Conference presentations in report period: Y. Tao, A. Yeckel, and J.J. Derby, “Modeling of Particle Engulfment in Microgravity Experiments,” 30th Annual Meeting of the American Society for Gravitational and Space Research, Pasadena, CA, October 25, 2014. Y. Tao, A. Yeckel, and J.J. Derby, “Particle Engulfment Dynamics during the Growth of Crystalline Silicon,” CSSC-8, 8th International Workshop on Crystalline Silicon for Solar Cells, Bamberg, Germany, May 5–8, 2015. Y. Tao, A. Yeckel, and J.J. Derby, “Particle Engulfment Dynamics during the Growth of Crystalline Silicon,” 20th American Conference on Crystal Growth and Epitaxy (ACCGE-20) and 17th U.S. Biennial Workshop on Organometallic Vapor Phase Epitaxy (OMVPE-17), Big Sky, MT, August 2–7, 2015. Sample Technical Results: We are preparing several papers for publication, and they are not yet ready for general distribution. However, we present a sample outcome from our initial work below. There are significant opportunities for increasing quality and reducing cost of crystalline silicon for solar applications via a better understanding of the growth process. One challenge is the formation of carbide and nitride precipitates in the melt that are engulfed by the solidification front to form inclusions in the mc-Si solid. These lower cell efficiency and can lead to wafer breakage and sawing defects. Minimizing the number of these engulfed particles will promote lower cost and higher quality silicon and will advance progress in commercial solar cell production. The primary goal of this research project is to understand the mechanisms that determine the dynamics of particle transport in silicon melts and the pushing or engulfment of these particles at a moving solid-liquid interface, specifically small silicon carbide (SiC) and silicon nitride particles (Si3N4) during the solidification of silicon. This work will coordinate with experiments conducted by the ParSiWal (Partikeleinfang bei der Siliziumkristallisation im Weltall) project funded by the German DLR and involving the Fraunhofer IISB, University of Freiburg, and University of Bayreuth. To better understand the physical mechanisms responsible for such inclusions during crystal growth, we have developed finite-element, moving-boundary analyses to assess particle dynamics during engulfment via solidification fronts. Two-dimensional, steady-state and dynamic models are developed using the Galerkin finite element method and elliptic mesh generation techniques in an arbitrary Eulerian-Lagrangian (ALE) implementation. This numerical approach allows for a rigorous representation of forces and dynamics previously inaccessible by approaches using analytical approximations.

	With this model, we have been able to address the critical factors that determine the transition from the stable pushing of a particle by a solid-melt interface to conditions where engulfment is inevitable. Theoretical predictions will be compared to ParSiWal experiments designed to probe the effects of particle engulfment during silicon solidification under terrestrial conditions in the laboratory and under microgravity conditions attained during TEXUS sounding rocket missions. Of particular interest and relevance is the discovery that oscillating solidification fronts, as would arise due to turbulent fluctuations in large-scale silicon melts, can drive engulfment of silicon carbide particles at average growth rates far below those predicted by prior analyses that assume constant growth conditions. This heretofore unexamined mechanism is a purely dynamic effect and may provide an explanation for the inability of classical, steady-state analyses to describe engulfment conditions in realistic growth systems.
Bibliography Type:	Description: (Last Updated: 09/04/2025)
Abstracts for Journals and Proceedings	Tao Y, Yeckel A, Derby JJ. "Modeling of Particle Engulfment in Microgravity Experiments." 30th Annual Meeting of the American Society for Gravitational and Space Research, Pasadena, CA, October 22-26, 2014. 30th Annual Meeting of the American Society for Gravitational and Space Research, Pasadena, CA, October 22-26, 2014. , Oct-2014
Abstracts for Journals and Proceedings	Tao Y, Yeckel A, Derby JJ. "Particle Engulfment Dynamics during the Growth of Crystalline Silicon." CSSC-8, 8th International Workshop on Crystalline Silicon for Solar Cells, Bamberg, Germany, May 5–8, 2015. CSSC-8, 8th International Workshop on Crystalline Silicon for Solar Cells, Bamberg, Germany, May 5–8, 2015. , May-2015